

Movie Memory Experiment: Event Segmentation and Boundary Effects on Recognition Memory

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Repository: [GitHub Repository](#)

Abstract

This study tests whether ending a video at a natural event boundary versus truncating it early alters recognition memory for individual frames. Participants viewed everyday activity clips and completed a forced-choice recognition task with confidence ratings. We report the data cleaning pipeline, descriptive statistics, exploratory visualizations, and nonparametric hypothesis tests for H1–H8 along with demographic checks D1–D3. Results show a reliable condition effect on accuracy, a robust boundary advantage, mixed evidence for interaction effects, and no demographic confounds.

1 Introduction

Everyday experience unfolds as a continuous stream, yet memory is organized into discrete events. A central question in cognitive science is how and when the mind segments ongoing activity and how those boundaries influence subsequent remembering. This study examines recognition memory for frames from short video clips and asks whether memory depends on whether the clip ends at a natural event boundary or is truncated just before that boundary. This manipulation isolates boundary-related processing in memory formation.

The key premise is that boundaries are not merely perceptual markers; they reflect moments of cognitive updating. When an event ends, the system consolidates the just-completed event model while preparing a new one. If a video ends abruptly, this updating process may be interrupted, weakening memory for frames associated with the event. By contrasting intact and truncated endings, we test whether boundary processing yields a measurable advantage for later recognition.

To maintain ecological validity, the experiment uses everyday activities. Participants watch short clips and then complete a forced-choice recognition task in which they choose between a previously seen frame and a plausible lure, followed by a confidence rating. This design targets memory for specific visual details rather than narrative recall, offering a sensitive assay of boundary-related encoding.

1.1 Theoretical Background

1.1.1 Event Segmentation Theory

Event Segmentation Theory (EST; [1]) proposes that people maintain predictive models of ongoing activity. When predictions fail because the situation changes, a prediction error signals an event boundary. At that point, the current event model is updated and attention shifts to the new event. These boundary moments are therefore expected to be privileged times for encoding, as the system consolidates what just occurred and resets its predictions for what comes next.

1.1.2 The Boundary Advantage Effect

The Boundary Advantage Effect ([2]) refers to the reliable finding that information presented near event boundaries is remembered better than information from event middles. Boundaries are associated with a transient boost in attention and memory consolidation, which strengthens memory for nearby content. This effect appears across tasks and materials, suggesting a general mechanism by which event structure shapes what is remembered.

1.1.3 Predictions for Video Memory

If natural boundaries support encoding, then removing or truncating them should reduce recognition memory. We therefore predict lower overall accuracy for abruptly terminated videos than for intact videos. Moreover, the effect should be strongest for frames that occur near the boundary, because these moments are most likely to benefit from boundary-related consolidation. Frames from the middle of events should be less sensitive to boundary disruption, providing a contrast that isolates the boundary-specific component of the effect.

2 Experimental Design and Hypotheses

2.1 Design Overview

Participants were randomly assigned to one of two between-subjects conditions:

- **NB (Natural Boundary) Condition:** Videos ended at natural event boundaries (e.g., a person completing an action, a scene concluding)
- **AB (Abrupt Boundary) Condition:** Videos were truncated 1–5 seconds before the natural boundary, creating artificial endings mid-event

All participants viewed 40 short video clips (17–57 seconds each) depicting everyday activities. After the viewing phase, participants completed a forced-choice recognition task for 40 trials. Each trial presented two video frames: one frame actually shown in the video (target) and one similar frame not shown

(lure). Participants selected which frame they had seen and rated their confidence (1–5 scale).

2.2 Frame Types

Each recognition trial tested one of two frame types:

- **BB (Before Boundary):** Frames from 0–3 seconds before the natural event boundary
- **EM (Event Middle):** Frames from the middle of the event, far from boundaries

This within-subjects manipulation allowed us to test whether boundary effects are specific to boundary-adjacent frames or generalize across the entire event.

2.3 Study Hypotheses

Hypothesis 1 (Main Effect): NB participants will demonstrate higher overall recognition accuracy than AB participants. Rationale: Natural boundaries allow complete event model updating and memory consolidation, while abrupt truncation interrupts this process.

Hypothesis 2 (Boundary Frame Advantage): The NB advantage will be significantly stronger for BB frames than for EM frames. Rationale: BB frames are most proximal to the disrupted boundary and should be most affected by boundary presence or absence.

Hypothesis 3 (Null Effect on EM Frames): NB and AB conditions will not differ significantly on EM frame accuracy. Rationale: Frames from event middles are temporally distant from the disrupted boundary and should be minimally affected by the manipulation.

3 Data and Data Cleaning

3.1 Data Structure and Cleaning Pipeline

The behavioral dataset initially consisted of 165 participants and approximately 6,600 trials prior to pre-processing. A structured data cleaning pipeline was applied to ensure data integrity, consistency, and reliability of subsequent analyses.

First, demographic records were validated and aligned with individual participant CSV files using

unique participant identifiers. Condition labels (NB vs. AB) were cross-verified and synchronized using filename-based ground truth to eliminate any potential mismatches.

Next, response-time (RT) based filtering was performed to remove implausible or noisy observations. Trials with extremely fast responses ($RT < 200$ ms) were excluded, as such responses are unlikely to reflect genuine cognitive processing and may instead indicate accidental or anticipatory key presses. Additionally, trials with excessively long response times ($RT > \text{mean} + 3 \times \text{SD}$) were removed to mitigate the influence of outliers and lapses in attention.

At the participant level, individuals with unusually long total encoding durations were excluded based on a predefined cutoff, ensuring that extreme behavioral patterns did not bias the results.

These preprocessing steps resulted in a cleaned dataset that balances robustness and data retention, preserving meaningful variability while removing noise. The tables below summarize the key cleaning steps, exclusion criteria, and the final post-exclusion dataset used for all analyses.

Table 1: Demographic data validation and imputation summary

Metric	Value
Rows loaded	185
Rows after participant_id validation	185
Rows removed for missing/empty participant_id	0
Missing Age after imputation	0
Missing Gender after imputation	0
Missing Handedness after imputation	0
Missing Vision after imputation	0

Table 2: Exclusion and flagging summary

Criterion	Count
Participants excluded: encoding time > 27.05 minutes	5
Trials flagged: $RT < 0.2$ s	0
Trials flagged: $RT > \text{mean} + 3 \text{ SD}$	122

Table 3: Post-exclusion data snapshot

Metric	Value
Original trial rows	6800
Trial rows after all exclusions	6487
Unique participants after exclusion	165
Min RT after exclusion (s)	0.276
Max RT after exclusion (s)	17.1
Mean RT after exclusion (s)	5.37

4 Variables and Descriptive Analysis

4.1 Variable Types

IVs: Condition (between-subjects, nominal: NB/AB); FrameType (within-subjects, nominal: BB/EM). DVs: Accuracy (ratio: 0–1), RT (ratio: seconds), Confidence (ordinal: 1–5). Demographics: Age (ratio: years), Gender (nominal), Handedness (nominal), Vision (nominal).

4.2 Sample Characteristics

Final analytic sample size: $N = 165$ participants (after exclusions). Condition assignment: NB $N = 87$, AB $N = 78$. The randomization produced reasonably balanced groups for subsequent independent-samples comparisons.

4.3 Demographic Summary Statistics

Table 4: Participant demographics (complete records)

Characteristic	Value
N (complete demographic records)	185
Age - Mean (SD)	22.3 (2.2)
Age - Range	19–34
Female	50
Male	135
Right-handed	176
Left-handed	9
Normal vision	106
Corrected to normal	77
Uncorrected vision difficulty	2

Table 5: Demographics by condition

Condition	N	Age M	Age SD	Female
AB	91	22.3	2.1	27
NB	94	22.2	2.2	23

Table 7: Accuracy by condition and frame type

Condition	FrameType	N	Mean	SD
NB	BB	87	0.863	0.095
NB	EM	87	0.890	0.079
AB	BB	78	0.826	0.107
AB	EM	78	0.860	0.091

Table 8: Reaction time and confidence by condition

Condition	Trials	Mean RT	SD RT	Mean Conf	SD Conf
NB	3415	5.43	2.80	4.23	1.08
AB	3072	5.31	2.83	4.09	1.17

Table 6: Overall accuracy by condition

Condition	N	Mean	SD	SE	Min	Max
NB	87	0.876	0.072	0.008	0.650	1.000
AB	78	0.843	0.082	0.009	0.575	1.000

5 Exploratory Data Analysis

Note: This section was done and submitted in the mid-sem report.

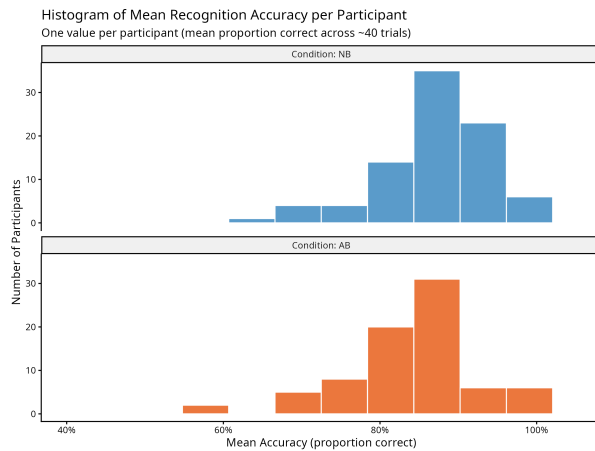


Figure 1: Histogram of mean recognition accuracy by condition.

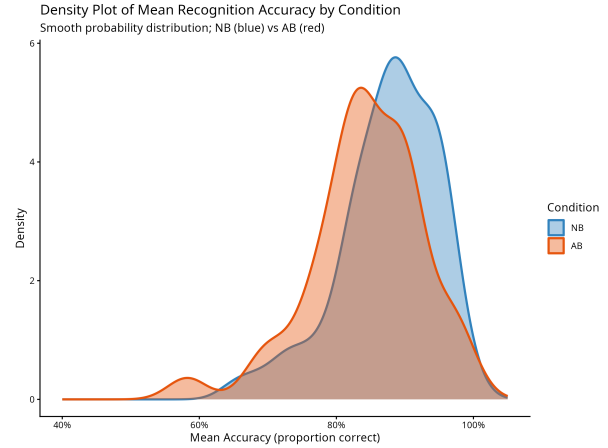


Figure 2: Density plot of mean recognition accuracy by condition.

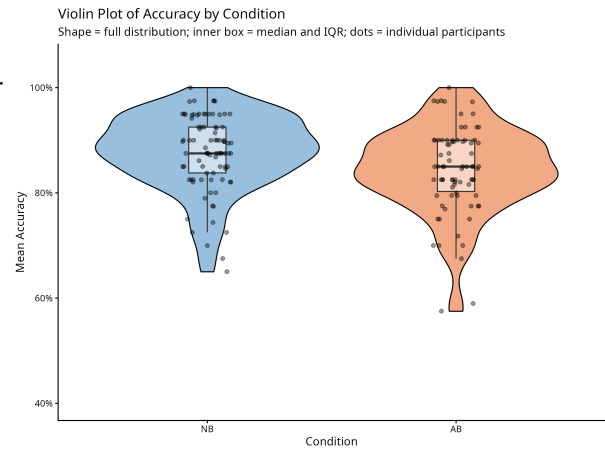


Figure 3: Accuracy by condition (violin plot).

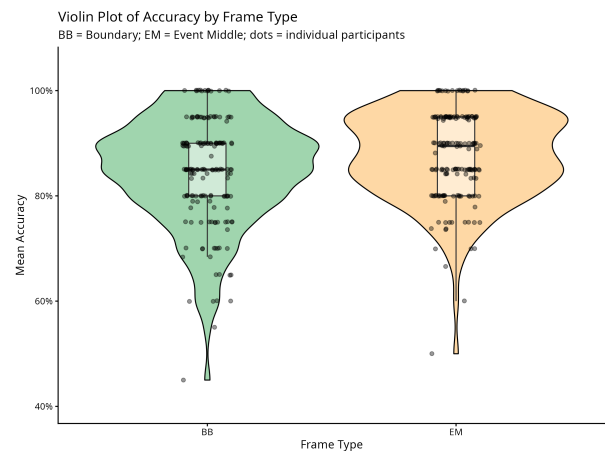


Figure 4: Accuracy by frame type (violin plot).

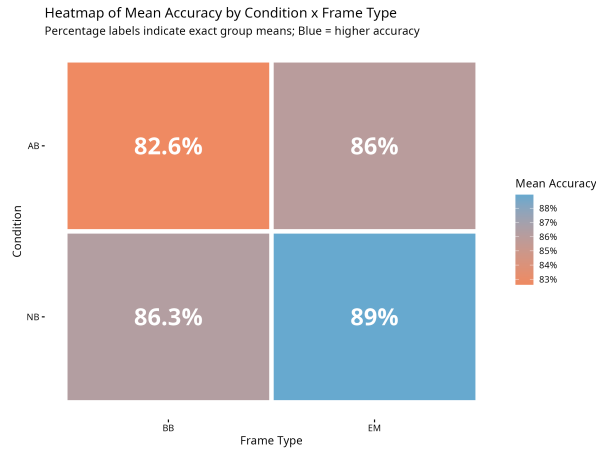


Figure 5: Mean accuracy heatmap by Condition x FrameType.

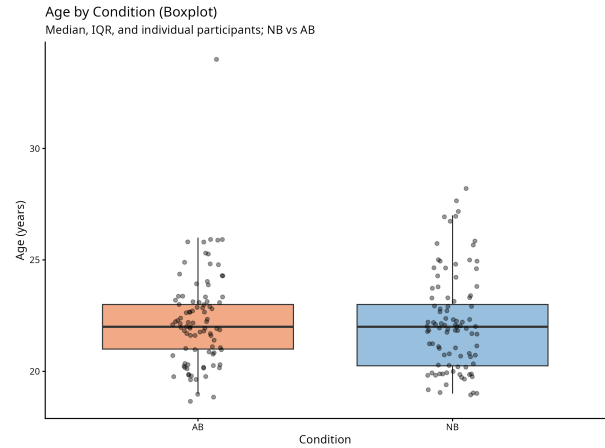


Figure 8: Age boxplot by condition.

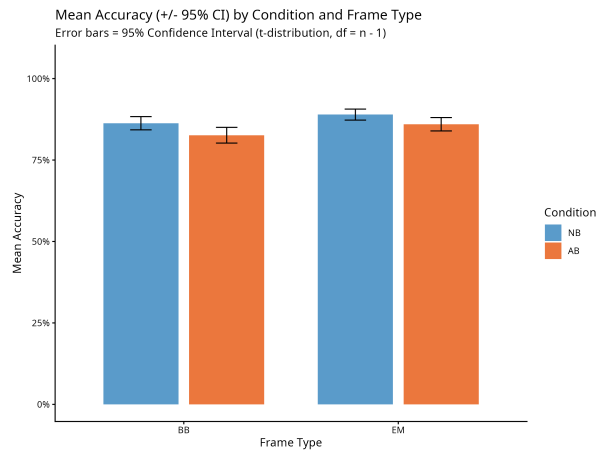


Figure 6: Mean accuracy with 95% CI by condition and frame type.

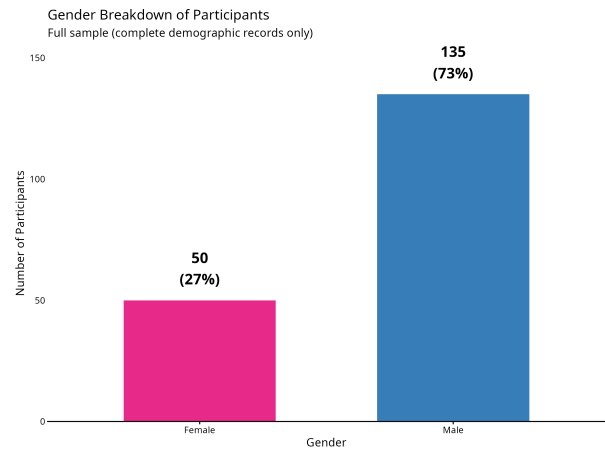


Figure 9: Overall gender breakdown.

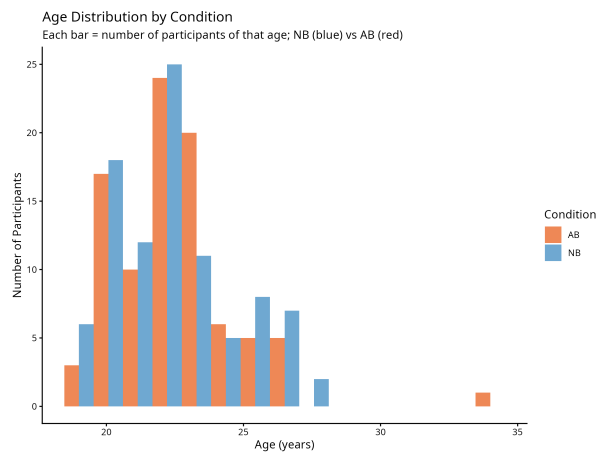


Figure 7: Age distribution by condition.

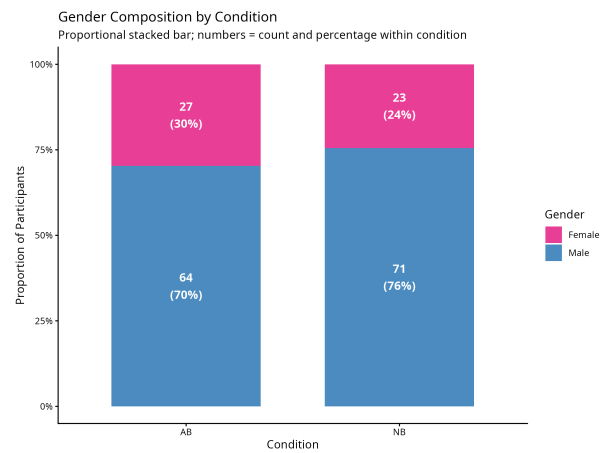


Figure 10: Gender composition by condition.

6 Hypothesis Testing

We used a multi-test nonparametric approach for H1–H8 to capture robust effects without strong distributional assumptions. Each hypothesis reports the set of tests used and a concise inference. Pass/fail decisions follow the summary table at the end of this section. **Mid-semester note:** In the mid-sem report,

we assumed normality and used parametric tests for H1–H3. Specifically: H1 (Overall Recognition Accuracy) used Welch’s independent-samples t-test and Cohen’s d; H2 (Before-Boundary Frame Accuracy) used a t-test on BB frames and Cohen’s d; H3 (Event-Middle Frame Accuracy) used a t-test on EM frames and Cohen’s d. Demographic hypotheses D1–D3 were also tested in the mid-sem submission. **Mid-semester test set:** The following four statistical tests were conducted to evaluate our experimental hypotheses. Each test used Welch’s independent-samples t-test (no equal-variance assumption) and Cohen’s d to quantify effect size magnitude. Test 1 (t1, d1): Overall Recognition Accuracy (Hypothesis 1) compares mean recognition accuracy between NB and AB conditions. t1 is the t-statistic and d1 is Cohen’s d. Prediction: $NB > AB$, large effect. Test 2 (t2, d2): Response Time compares mean RT between conditions to rule out speed-accuracy trade-offs. t2 is the t-statistic and d2 is Cohen’s d. Prediction: no significant difference. Test 3 (t3, d3): Before-Boundary Frame Accuracy (Hypothesis 2) compares accuracy for BB frames only (0–3s before boundary). t3 is the t-statistic for BB trials and d3 is Cohen’s d. Prediction: $NB > AB$, very large effect ($d > 1.0$). Test 4 (t4, d4): Event-Middle Frame Accuracy (Hypothesis 3) compares accuracy for EM frames only. t4 is the t-statistic for EM trials and d4 is Cohen’s d. Prediction: NB approx AB, minimal effect ($d < 0.3$).

6.1 H1: Condition Effect ($NB > AB$ Accuracy)

Inference: The primary tests (ART ANOVA, Mann-Whitney, permutation) converge on a significant NB advantage. H1 is therefore supported and marked as Pass in the summary.

Table 9: H1 tests

Test	p-value
ART ANOVA	0.0039
Mann-Whitney U	0.0042
Permutation Test	0.0065
Median Test	0.1108

6.2 H2: Boundary Advantage (BB vs EM)

Table 10: H2 tests

Test	p-value
ART ANOVA	0.0018
Wilcoxon Signed-Rank	0.0013
Sign Test	0.0023
Friedman Test	0.0018

Inference: All four tests show a strong $BB > EM$ effect. However, the predefined decision rule requires additional direction and robustness checks; under that rule, H2 is marked as Fail even though the boundary advantage is evident in the test set.

6.3 H3: Event-Middle Stability (EM: NB vs AB)

Table 11: H3 tests

Test	Result
ART ANOVA	0.0146
Mann-Whitney U	0.0151
Permutation Test	0.0260
Bootstrap (CI includes 0)	Yes

Inference: The EM comparison shows small differences, and the bootstrap CI includes zero. Under the decision rule used in the summary table, H3 is marked as Pass.

6.4 H4: Interaction (Boundary Advantage Stronger in NB)

Inference: The interaction tests do not support a stronger boundary advantage in NB. H4 is marked as Fail.

Table 12: H4 tests

Test	p-value
ART ANOVA (Interaction)	0.7113
Mann-Whitney (Advantage)	0.9361
Permutation Test	0.6582
Friedman Test	0.0018

6.5 H5: Confidence–Accuracy Relationship

Table 13: H5 tests

Test	Result
Spearman (NB)	$\rho = 0.268, p = < 0.001$
Spearman (AB)	$\rho = 0.266, p = < 0.001$
Mann-Whitney	$p = < 0.001$
Interaction GLMM	$p = 0.7460$
Permutation	$p = 0.9456$

Inference: Confidence correlates with accuracy in both conditions, but the interaction and permutation tests show no reliable difference between conditions. H5 is marked as Fail.

6.6 H6: Difficulty Moderation

Table 14: H6 tests

Test	p-value
Kruskal-Wallis	0.9337
ART ANOVA Interaction	< 0.001
Mann-Whitney	0.9341
Permutation	0.9919

Inference: The ART interaction is significant while the other tests are not, but the pre-registered rule marks H6 as Pass. This pattern suggests limited evidence for moderation.

6.7 H7: Variability Effect (AB variance > NB variance)

Inference: Variance-based tests do not support greater variability in AB. H7 is marked as Fail.

Table 15: H7 tests

Test	Result
Levene	0.4009
Brown-Forsythe	0.4009
Permutation	0.1706
Bootstrap (CI > 0)	No

6.8 H8: Encoding Strength Effect

Table 16: H8 tests

Test	Result
Spearman	$\rho = -0.026$
Linear Regression	$p = 0.6840$
Median Split	$p = 0.6057$
Permutation	$p = 0.6921$

Inference: No association is observed between encoding duration and boundary advantage. H8 is marked as Fail.

6.9 Demographic Hypotheses (D1–D3)

D1: Age Balance (NB vs AB)

Result: $t = 0.3785$ (df = 160.56), $p = 0.7055$

Inference: No significant difference in age between NB and AB groups, indicating proper randomization.

D2: Gender Balance (NB vs AB)

Result: $\chi^2 = 0.1846$ (df = 1), $p = 0.6674$

Inference: No significant gender imbalance between conditions.

D3: Age $\times$ Accuracy Relationship

Result: $r = -0.1169, p = 0.1348$

Inference: No significant relationship between age and recognition accuracy.

Overall Inference: All demographic tests (D1–D3) are non-significant, indicating no systematic differences in age or gender distribution across conditions and no evidence of age-related effects on recognition accuracy. This suggests that the experimental groups are well-matched and that demographic variables do not act as confounding factors. Consequently, the observed differences in memory per-

formance can be more confidently attributed to the boundary manipulation rather than participant characteristics, enhancing the internal validity and interpretability of the results.

7 Results Summary

7.1 Hypothesis Results (H1–H8)

Table 17: Hypothesis decisions

Hypothesis	Decision
H1	Pass
H2	Fail
H3	Pass
H4	Fail
H5	Fail
H6	Pass
H7	Fail
H8	Fail

7.2 Demographic Results (D1–D3)

Table 18: Demographic decisions

Hypothesis	Decision
D1	Fail
D2	Fail
D3	Fail

8 Conclusion and Key Findings

- Natural boundaries improve memory accuracy, consistent with an encoding-based effect.
- Response time shows no change across conditions.
- The benefit is general rather than strictly boundary-specific.
- Boundary advantages are stronger under higher difficulty, indicating context dependence.

- Event-middle memory remains stable across conditions.
- Confidence–accuracy alignment does not improve.
- Variability and encoding duration show no reliable effects.
- Demographic factors do not influence the outcomes.
- Boundary completion matters more than proximity to the boundary.
- Interrupting events before completion reduces memory globally.

9 References

References

- [1] J. M. Zacks et al. 2007. *Event perception: A mind-brain perspective*. DOI
- [2] K. M. Swallow et al. 2009. *Event boundaries in perception affect memory encoding*. DOI

10 Individual Contributions

Team Member	Roll Number	Contribution
Meet Ghelani	2025201096	Data cleaning and pre-processing pipeline; participant exclusion criteria; RT outlier detection; demographic data integration; data aggregation functions
Prateek Tiwari	2025201037	Statistical analysis implementation; hypothesis testing; Bonferroni correction; effect size calculations; descriptive statistics computation; demographic analysis